

1. Continuous concentration problem (§3)

$$\mathcal{T}\psi = \lambda \psi, \quad \mathcal{T} = B_R B_K B_R$$

on $L^2([0, R], r dr)$, fixed angular order n

CPSWFs $\psi^{(m)}$; eigenvalues $\lambda_m^n(c)$, $c = KR$

2. Fourier-Bessel bridge (§4.1)

$$\varphi_{n,k}(r) = \frac{\sqrt{2}}{R} J_n(j_{n,k} r/R) / J_{n+1}(j_{n,k})$$

$k = 1, \dots, P$ (closure sizing $c = j_{n,P+1}$ optional)

$$V_P = \text{span}\{\varphi_{n,1}, \dots, \varphi_{n,P}\}$$

3. Finite radial prolate matrix (§4.2-4.3)

$$(B_K^{(B)})_{mk} = 2 j_{n,m} j_{n,k} \int_0^c \frac{u J_n(u)^2 du}{0(u^2 - j_{n,m}^2)(u^2 - j_{n,k}^2)}$$

real symmetric, PSD, eigenvalues in $[0, 1]$

depends only on (n, c, P)

eigendecomposition

4. Finite radial prolate sequences (§4.4)

$$B_K^{(B)} c^{(m)} = \lambda_m^{(P)} c^{(m)}$$

orthonormal eigenvectors $c^{(m)}$

$\lambda_0^{(P)} \geq \lambda_1^{(P)} \geq \dots$: plateau – plunge – tail

*Galerkin
compression*

5. Finite radial prolate functions (§4.5)

$$\psi_P^{(m)}(r) = \frac{\sqrt{2}}{R} \sum_{k=1}^P \frac{c_k^{(m)}}{J_{n+1}(j_{n,k})} J_n(j_{n,k} r/R)$$

synthesis S_P : exact isometry $\mathbb{R}^P \cong V_P$

doubly orthogonal family

synthesis S_P

$P \rightarrow \infty$

6. Approximation of CPSWFs (§5, Thms 5-7)

$$0 \leq \lambda_m - \lambda_m^{(P)} \leq 2\lambda_0 \delta_P(E_{m+1})^2 \quad (\text{Thm 6})$$

leading-mode subspace $\rightarrow$ CPSWFs in weighted L^2 (Thm 7a)

$$\|\mathcal{T}_P - \mathcal{T}\| \rightarrow 0 \quad (\text{Thm 5})$$

approximates